Gurjot Singh

Software Engineer Intern Candidate | B.Tech 2027

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www.gurjot.codes | GitHub | LinkedIn | LeetCode | Codeforces

Summary

Final-year B.Tech student (Class of 2027, CGPA 8.29) with strong fundamentals in data structures, algorithms, and system design, backed by Codeforces Specialist rank and 750+ LeetCode solutions. Experienced in building secure, high-throughput backend systems (Golang, Python, Node.js) and in engineering enterprise AI agents with RAG pipelines at Accenture. Passionate about security, cyber resilience, and performance bottlenecks. **Available onsite in Bangalore, January–May 2027.**

Education

National Institute of Technology (NIT), Jalandhar
B.Tech. in Data Science and Engineering | Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Systems Design, ML, N

Gobindgarh Public School, Mohali 2021 – 2023
Class 11–12 (Science Stream) *89%*

Skills

Programming Languages: Python, C++, Golang, JavaScript, TypeScript

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, System Design, Object-Oriented Programming (OOP)

Security & Systems: Applied Cryptography (RSA, AES-256, HMAC, SHA-256, Double Ratchet), E2EE, RBAC, Multi-Factor Auth, WebSockets, gRPC, Protocol Buffers

Backend, Cloud & Databases: Node.js, Express.js, Flask, REST APIs, GraphQL, tRPC, PostgreSQL, MongoDB, MySQL, Redis, Docker, Kubernetes, AWS, Linux, Git

AI & Agentic Systems: LLMs, RAG Systems, Agentic AI, LangChain, LangGraph, Vector Databases, TensorFlow, Scikit-learn

Frontend: React.js, Next.js, Tailwind CSS

Experience

- **Advance Application Engineering (AEH) Intern — AI Agent Development — Accenture, Bangalore — Jun 2026–Jul 2026**:
  - Architected enterprise **AI agents** using **RAG** pipelines to ground LLMs in multi-tenant data, eliminating context drift.
  - Engineered semantic search infrastructure using **vector embeddings**, boosting context retrieval precision by **40%**.
  - Automated document ingestion in **Python** and built **React.js** dashboards to monitor agent token usage and latency in real time.
- **CTP (NITJ Placement Portal)** [Live Site] — **Backend Lead — NIT Jalandhar — 2024–2026**:
  - Owned the **Node.js/Express.js** backend serving **1,200+ users** across **25+ REST APIs** for end-to-end recruitment workflows.
  - Diagnosed query bottlenecks and restructured **MongoDB** indexing, cutting data retrieval time by **35%** under **10k+ daily requests**.
  - Implemented granular **RBAC** for secure access control and shipped features in **Agile/Scrum** sprints, accelerating shortlisting cycles by **20%**.
- **Third Space Travel (Travel Booking Platform)** [Live Site] — **Full-Stack Developer — On-Site — 2026**:
  - Built a production platform with **React**, **TypeScript**, **tRPC**, and **Prisma/MySQL**; optimized queries to cut latency by **50ms**.
  - Integrated **PhonePe** payments (**99.9% success rate**, **20%** fewer drop-offs) and deployed on **AWS EC2** with **Docker**, reaching a **95+ Lighthouse** score.

Projects

- **Shadow-Room: E2EE Secure Chat System — 2026** **Live Demo** | **Repo**:
  - Built an end-to-end encrypted messaging platform in **Golang** and **React.js** with **WebSockets** for low-latency, real-time group communication.
  - Implemented cryptographic handshakes (**RSA, HMAC, SHA-256**) and the **Double Ratchet Algorithm** for forward secrecy and message integrity, scaling groups via Sender Keys.
  - Added multi-factor authentication and **Protocol Buffers** serialization to support high-throughput concurrent sessions.
- **Secure Token Allocation System — NIT Jalandhar — 2026** **Live Demo** | **Repo**:
  - Developed a cryptographically secure token generation and tracking system to automate physical reporting for **1,200+ incoming students**.
  - Used **AES-256** encryption and secure hashing to prevent token forgery and keep the queue tamper-proof.
  - Built a responsive **React.js** dashboard for live allocations, reducing wait times by **60%**.
- **DataPulse: Operations Management Hub — 2025** **Live Demo** | **Repo**:
  - Engineered a data operations API with **Flask** and **GraphQL**; custom resolvers eliminated over-fetching and cut frontend payload sizes by **70%**.
  - Added **Redis** caching over **PostgreSQL** to sustain a **200ms** response time on high-frequency queries.

Achievements & Leadership

- **Codeforces & LeetCode:** Achieved **Specialist** rank on Codeforces (Peak Rating: **1502**, Global Rank **676** in Div. 2); ranked in the **Top 5%** globally on LeetCode with **750+** problems solved (**1,050+** across all platforms).
- **Speed Quant (UTKANSH'25) — First Place:** Secured **1st Rank** out of **200+** participants in NIT Jalandhar's flagship quantitative competition, solving high-pressure mathematical and algorithmic problems.
- **AWS Club (NIT Jalandhar) — Technical Lead:** Led hands-on bootcamps on **Cloud Architecture**, **DevOps**, and **GenAI**, mentoring **300+** students in scalable backend systems and containerization.